UNIT I: HTML5 & JavaScript Essentials

HTML5: Structure, Forms, Tables, Media, Security Practices. JavaScript: Syntax, Functions, DOM Manipulation, Async Programming.

Tasks:

1. **Create a static web page** using HTML5 elements (headings, images, lists, forms).
2. **Add validation logic** to the HTML form using JavaScript functions.
3. **Build an interactive component** with DOM manipulation and event handling.
4. **Fetch data from an API** using JavaScript Fetch API and display it on the page.

Task-2:

Aim: Add validation logic to the HTML form using JavaScript functions

Task2u1.html:

```
<html>
<head>
<title>Form Validation</title>
<script>
function validateForm() {
let name = document.getElementById("name").value;
let roll = document.getElementById("roll").value;

if (name.length < 5) {
alert("Name must contain at least 5 characters.");
return false;
}

if (roll.length < 3) {
alert("Roll Number must contain at least 3 characters.");
return false;
}

alert("Form Submitted Successfully!");
alert("Name: " + name + "\nRoll Number: " + roll);

console.log("Name: " + name + "\nRoll Number: " + roll);

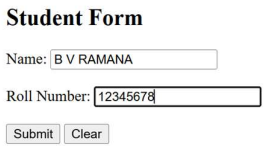
document.getElementById("result").innerHTML =
"<h3>Student Details</h3>" +
"Name: " + name + "<br>Roll Number: " + roll;
return false;
}
function clearForm() {
document.getElementById("name").value = "";
document.getElementById("roll").value = "";
document.getElementById("result").innerHTML = "";
}
```

```
</script>
</head>
<body>
<h2>Student Form</h2>
<form onsubmit="return validateForm()">
Name:
<input type="text" id="name"><br><br>
Roll Number:
<input type="text" id="roll"><br><br>
<input type="submit" value="Submit">
<input type="button" value="Clear" onclick="clearForm()">
</form>
<br>
<div id="result"></div>

</body>
</html>
```

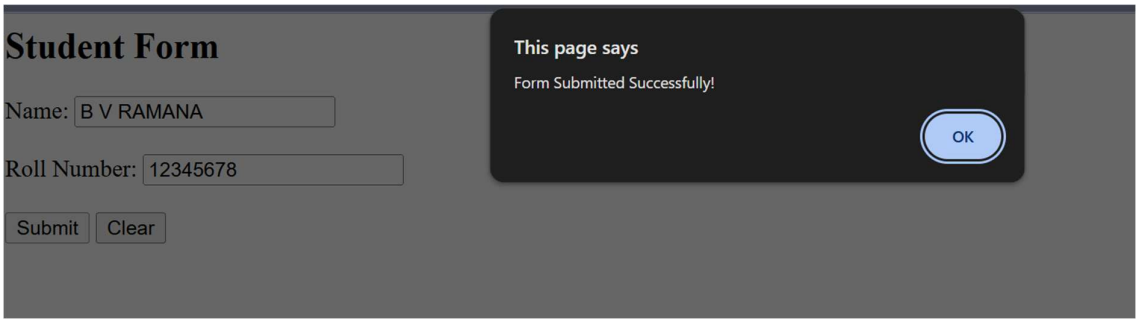
OUTPUT:

Step-1

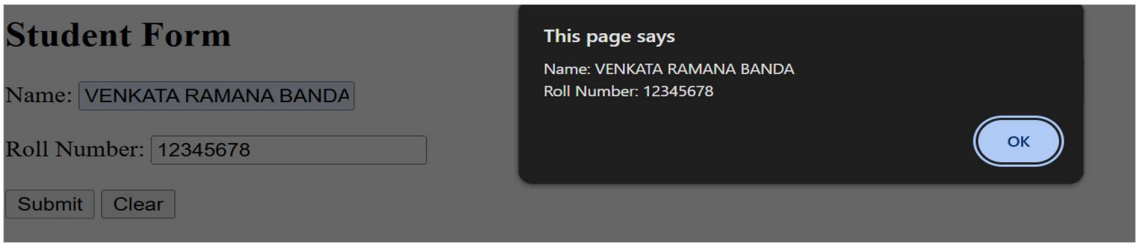


After Submitted

Stee-2

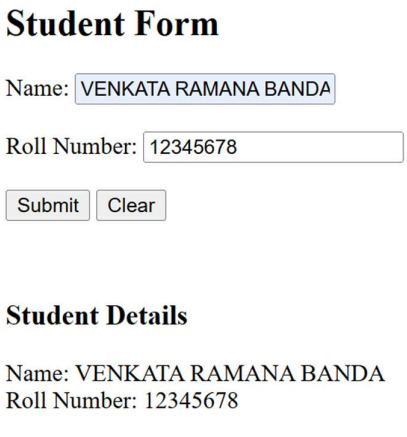


Click on OK



Click on OK

Step-3



Task-3:

Aim: Build an interactive component with DOM manipulation and event handling.

DOM stands for **Document Object Model**.

Definition:

The **Document Object Model (DOM)** is a programming interface that represents an HTML document as a tree of objects. It allows JavaScript to access and modify the content, structure, and style of a web page.

DOM (Document Object Model) is a programming interface that allows JavaScript to access, modify, add, or remove HTML elements dynamically.

Common DOM Methods:

- document.getElementById() – Selects an element by its ID.
- document.getElementsByTagName() – Selects elements by tag name.
- document.getElementsByClassName() – Selects elements by class name.
- document.querySelector() – Selects the first matching element.
- document.querySelectorAll() – Selects all matching elements.

document.getElementById("id")
document.getElementsByClassName("class")
document.getElementsByTagName("tag")
document.querySelector("selector")
document.querySelectorAll("selector")

task3u1.html:

```
<html>
<head>
<script>
function hideText() {
 document.getElementById("msg").style.display = "none";
}
function showText() {
document.getElementById("msg").style.display = "block";
}
</script>
</head>
<body>
<p id="msg"><b>This is Task-3 in UNIT-I</b></p></br>
<button onclick="hideText()">Hide</button>
<button onclick="showText()">Show</button>
</body>
</html>
```

OUTPUT:

This is Task-3 in UNIT-I

Hide Show

- **Click on Hide: output like this**

Hide Show

Task-4:.

Aim: Fetch data from an API using JavaScript Fetch API and display it on the page.

An **API (Application Programming Interface)** is a way for two software applications to communicate and exchange data.

JSON (JavaScript Object Notation) is a lightweight data format used to store and exchange data between a server and a web application.

- **API** → Provides the data.
- **JSON** → The format in which the data is sent.
- **Fetch API** → Retrieves the JSON data in JavaScript.

Procedure:

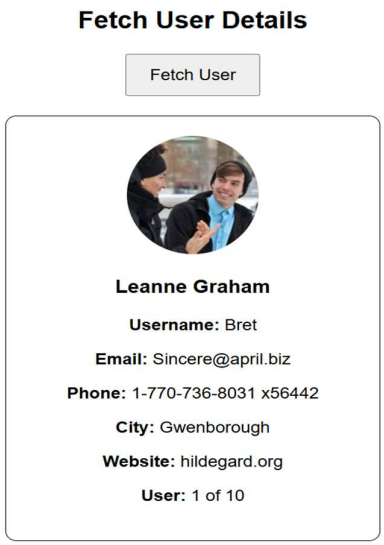
1. Use a public API (e.g., https://jsonplaceholder.typicode.com/users or "https://randomuser.me/api/")).
2. Write JavaScript fetch() to retrieve data.
3. Convert response to JSON.
4. Display Outputs in the DOM.

task4u1.html:

```
<html>
<head>
<title>Fetch User One by One</title>
<style>
body {
font-family: Arial;
text-align: center;
}
#result {
width: 300px;
margin: 20px auto;
padding: 20px;
border: 1px solid black;
border-radius: 10px;
}
img {
width: 120px;
height: 120px;
border-radius: 50%;
}
button {
padding: 10px 20px;
font-size: 16px;
}
</style>
</head>
<body>
<h2>Fetch User Details</h2>
<button onclick="fetchUser()">Fetch User</button>
<div id="result"></div>
<script>
let users = [];
let index = 0;
function fetchUser() {
if (users.length == 0) {
fetch("https://jsonplaceholder.typicode.com/users")
.then(response => response.json())
.then(data => {
users = data;
displayUser();
```

```
})
.catch(error => {
document.getElementById("result").innerHTML =
"Error: " + error;
});
} else {
displayUser();
}
}
function displayUser() {
if (index >= users.length) {
document.getElementById("result").innerHTML =
"<h3>No more users!</h3>";
return;
}
let user = users[index];
document.getElementById("result").innerHTML =
"<img src='https://i.pravatar.cc/150?img=" + (index + 1) + "'>" +
"<h3>" + user.name + "</h3>" +
"<p><b>Username:</b> " + user.username + "</p>" +
"<p><b>Email:</b> " + user.email + "</p>" +
"<p><b>Phone:</b> " + user.phone + "</p>" +
"<p><b>City:</b> " + user.address.city + "</p>" +
"<p><b>Website:</b> " + user.website + "</p>" +
"<p><b>User:</b> " + (index + 1) + " of " + users.length + "</p>";
index++;
}
</script>
</body>
</html>
```

OUTPUT:



Click on fetch user (get one more)

